



BUSINESS STATISTICS (BUS 215)

TOPIC 4

MEASURES OF DISPERSION

LECTURER: MR ELISA EBIYAMU

BUSINESS STATISTICS (BUS 215)

Online Lesson 4.2

- 1. Variance of Grouped Data,**
- 2. Coefficient of Variation**

VARIANCE AND STANDARD DEVIATION OF GROUPED DATA

Introduction

- In lesson 4.1, we were able to compute variance (σ^2) and standard deviation from a set of data using the following definition formula.

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N} \text{ for population variance}$$

$$s^2 = \frac{\sum (X - \bar{X})^2}{n-1} \text{ for sample variance}$$

- We could also use computation formulas for variance

$$s^2 = \frac{\sum x^2 - \left(\frac{\sum x}{n}\right)^2}{n-1} \text{ or } s^2 = \frac{n \sum x^2 - (\sum x)^2}{n(n-1)}$$

Introduction

- You may recall that statisticians group data especially when there are many observations and the range is wide.
- In topic 3, we estimated the mean of grouped data where by the class mid point was used in place of actually observed values.
- In this section of lesson 4.2 we will be able to estimate the variance and standard deviation of grouped data.

Variance and Standard; Deviation (s) of Grouped Data

- The procedure for finding the variance and standard deviation for grouped data is similar to that for finding the mean for grouped data, and it uses the midpoints of each class.
- This procedure uses the shortcut (Computation) formula, and
 - X_m is the symbol for the class midpoint
 - f is frequency of each class.

$$s^2 = \frac{n(\sum f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}$$

$$s = \sqrt{\frac{n(\sum f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}}$$

Variance and Standard Deviation (s) of Grouped Data

Procedure

Step 1: Make a table as shown below

A Class	B Frequency	C Midpoint	D $f \cdot X_m$	E $f \cdot X_m^2$

Step 2: Multiply the frequency by the midpoint for each class, and place the products in column D.

Step 3: Multiply the frequency by the square of the midpoint, and place the products in column E.

Variance and Standard Deviation (s) of Grouped Data

Step 4: Find the sums of columns B, D, and E.

The sum of column B is n .

The sum of column D is $\sum f \cdot X_m$

The sum of column E is $\sum f \cdot X_m^2$

Step 5: Substitute in the formula and solve to get the variance.

$$s^2 = \frac{n(\sum f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}$$

Step 6: Take the square root to get the standard deviation.

Variance and Standard Deviation (s) of Grouped Data

Example

- Find sample variance and sample standard deviation for the frequency distribution for data of miles that 20 runners ran during one week.

Class	Frequency
5.5 – 10.5	1
10.5 – 15.5	2
15.5 – 20.5	3
20.5 – 25.5	5
25.5 – 30.5	4
30.5 – 35.5	3
35.5 – 40.5	2

Variance and Standard Deviation (s) of Grouped Data

A Class	B Frequency	C Midpoint	D $f \cdot X_m$	E $f \cdot X_m^2$
5.5–10.5	1	8	8	64
10.5–15.5	2	13	26	338
15.5–20.5	3	18	54	972
20.5–25.5	5	23	115	2,645
25.5–30.5	4	28	112	3,136
30.5–35.5	3	33	99	3,267
35.5–40.5	2	38	76	2,888
	<u>20</u> $n = 20$		<u>490</u> $\Sigma f \cdot X_m = 490$	<u>13,310</u> $\Sigma f \cdot X_m^2 = 13,310$

Variance and Standard Deviation (*s*) of Grouped Data

$$\text{variance} = s^2 = \frac{n(\sum f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}$$

$$s^2 = \frac{20(13,310) - (490)^2}{20(20 - 1)}$$

$$= \frac{26,100}{380} = 68.7$$

$$\text{Standard deviation} = \sqrt{68.7}$$

$$= 8.3$$

Uses of Variance and Standard Deviation

1. Determining the spread of the data.

If the variance or standard deviation is large, the data are more dispersed. This information is useful in comparing two or more variables.

2. Determining consistency of a variable. The variance and standard deviation are used to determine the number of data values that fall within a specified interval in a distribution.
3. Used to compute standard error of a statistic in inferential statistics .

Approximating Standard Deviation

- The range can be used to approximate the standard deviation. The approximation is called the **range rule of thumb**.

$$s \approx \frac{\text{range}}{4}$$

Example

- For the data set 5, 8, 8, 9, 10,
 - The range is $13 - 5 = 8$.
 - Then $s \approx \frac{8}{4} \approx 2$

Approximating Standard Deviation

- However, the actual standard deviation is 2.7.
- The range rule of thumb underestimates the standard deviation in this case
- The range rule of thumb is only an *approximation* and should be used when the distribution of data values is unimodal and roughly symmetric

Approximating Standard Deviation

- The range rule of thumb can be used to estimate the largest and smallest data values of a data set.
- The smallest data value will be approximately 2 standard deviations below the mean.
- The largest data value will be approximately 2 standard deviations above the mean of the data set.
- The mean and standard deviation for the previous data set are 9.3 and 2.7 respectively. hence,

$$\text{Smallest data value} = \bar{X} - 2s = 9.3 - 2(2.7) = \mathbf{3.9}$$

$$\text{Largest data value} = \bar{X} + 2s = 9.3 + 2(2.7) = \mathbf{14.7}$$

Practice question

- The table below shows data of ages of students at a college in Malawi.

Age in years	Number of
15 up to 20	43
20 up to 25	88
25 up to 30	94
30 up to 35	78
35 up to 40	29

- Estimate mean age and standard deviation

COEFFICIENT OF VARIATION

Coefficient of Variation

- Whenever two samples have the same units of measure, the variance and standard deviation for each can be compared directly.
- For example, suppose an automobile dealer wanted to compare the standard deviation of miles driven for the cars she received as trade-ins on new cars. She found that for a specific year, the standard deviation for Buicks was 422 miles and the standard deviation for Cadillacs was 350 miles.
- She could say that the variation in mileage was greater in the Buicks.
- But what if a manager wanted to compare the standard deviations of two different variables, such as the number of sales per salesperson over a 3-month period and the commissions made by these salespeople?
- A statistic that allows you to compare standard deviations when the units are different, as in this example, is called the *coefficient of variation*.

Coefficient of Variation

- The **coefficient of variation**, denoted by CVar, is the standard deviation divided by the mean.
- The result is expressed as a percentage.

For samples,

$$\text{CVar} = \frac{s}{\bar{X}} \cdot 100$$

For populations,

$$\text{CVar} = \frac{\sigma}{\mu} \cdot 100$$

Coefficient of Variation

Example

The mean for the number of pages of a sample of women's fitness magazines is 132, with a variance of 23; the mean for the number of advertisements of a sample of women's fitness magazines is 182, with a variance of 62. Compare the variations.

Coefficient of Variation

- Solution

$$CV_{\text{ar}} = \frac{\sqrt{23}}{132} \cdot 100 = 3.6\% \quad \text{pages}$$

$$CV_{\text{ar}} = \frac{\sqrt{62}}{182} \cdot 100 = 4.3\% \quad \text{advertisements}$$

- The number of advertisements is more variable than the number of pages since the coefficient of variation is larger for advertisements

Coefficient of Variation

Practice Question

The lengths (in feet) of the main span of the longest suspension bridges in the United States and the rest of the world are shown below. Which set of data is more variable?

United States	: 4205, 4200, 3800, 3500, 3478, 2800, 2800, 2310
World	: 6570, 5538, 5328, 4888, 4626, 4544, 4518, 3970

THANK YOU